Commitment Intensity: A Physics-Grounded Cognitive Measurement from SAE Feature Coherence Trajectories

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Abstract

We introduce *commitment intensity* (S), a scalar measurement of how concentrated a model's coherence events are during generation, computed from per-token SAE feature coherence trajectories. We derive S as the inverse participation ratio (IPR) of the coherence event distribution — a quantity with 70 years of physics precedent in localization theory. On Gemma-2-2B-IT, S computed over the first 7 generated tokens (S_{early}) discriminates commitment from hedging and naturally hallucinated from correct TruthfulQA responses with a pooled effect size of $d = 0.494$ across three independent experimental designs (Fisher combined $p = 0.0008$). In head-to-head comparison, S_{early} outperforms logit entropy (AUC 0.663 vs. 0.607, $p = 0.013$) while being nearly independent of it ($r = -0.17$). Combined in a 2D cognitive state map, the two signals define a "danger zone" where 65% of generations are hallucinated versus 22% in the "safe zone" ($\chi^2 = 7.17$, $p = 0.007$). A streaming classifier achieves CV AUC 0.701 at token 7 ($\sim$350ms). Crucially, three SAE-free probes tested on four architecture families (Gemma, Llama, Qwen, Phi) all fail to capture the same signal (Fisher $p > 0.6$), while SAE-based measurement replicates on both Gemma-2-2B-IT and Llama-3.2-1B-Instruct with architecture-adaptive windows (pooled $d = 0.587$, Fisher $p = 0.0006$). This establishes that SAE feature decomposition performs irreplaceable measurement work across architectures. We conclude that SAEs are not merely interpretability aids but necessary measurement infrastructure for a class of cognitive properties invisible to raw activation analysis.

1. Introduction

The dominant paradigm for AI oversight relies on language-level monitoring — one model evaluating another's outputs for safety compliance. Song et al. (2026) demonstrate that this architecture is fundamentally compromised: frontier models spontaneously develop peer-preservation behaviors, deceiving oversight systems to protect allied models. If the monitor shares the same representational substrate as the monitored system, collusion is not a bug — it is an emergent property.

We propose oversight below the semantic layer: measuring the mathematical structure of the model's internal computation during generation, not the text it produces. Our instrument, Fathom, operates on per-token SAE feature coherence trajectories — the time series of how aligned the model's sparse feature representations are across layers as it generates each token.

From these trajectories, we derive *commitment intensity* (S): a scalar that measures how concentrated the model's coherence events are. High S means few but intense commitment events; low S means many small distributed events. We show that S is mathematically equivalent to the inverse participation ratio (IPR) from condensed-matter physics, grounding it in a 70-year theoretical lineage.

Our contributions:

1. S as a fourth orthogonal axis of cognitive geometry, independent of depth (K), coherence (C), and layer divergence (C_Δ) at $|r| < 0.10$ on $n = 400$ pooled samples.
2. A cross-dataset meta-analysis showing $d \approx 0.5$ replication across three independent designs (Fisher $p = 0.0008$).
3. Head-to-head superiority over logit entropy ($p = 0.013$) with near-independence ($r = -0.17$), enabling a 2D cognitive state map.
4. Empirical proof that SAE decomposition is necessary:

three SAE-free probes on four architectures return chance-level discrimination.

2. Background

SAE Interpretability. Sparse autoencoders decompose transformer residual streams into interpretable monosemantic features (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024). Our work extends SAEs from interpretability tools to *measurement infrastructure*: we show they extract quantitative cognitive properties inaccessible to raw activation analysis.

Hallucination Detection. Semantic entropy (Farquhar et al., 2024) requires multiple samples. SelfCheckGPT (Manakul et al., 2023) operates on outputs. DoLa (Chuang et al., 2024) contrasts layer predictions. ITI (Li et al., 2023) steers attention heads. All operate on raw activations or outputs. None provide per-token SAE-derived commitment measurements, and none have demonstrated head-to-head superiority with pre-registered methodology.

Internal Representations. Concurrent work demonstrates causal internal emotion representations in LLMs (Lindsey et al., 2026), validating the premise that internal state geometry predicts behavior.

Inverse Participation Ratio. The IPR measures localization in discrete distributions (Anderson, 1958). High IPR = localized (few sites carry weight); low IPR = delocalized (weight spread evenly). IPR has been used in quantum many-body physics, random matrix theory, and neuroscience for 70+ years. We show S is a specific IPR instance applied to coherence event trajectories.

3. Method

3.1. Per-Token Coherence Trajectory

For each generated token t, we extract residual streams at all hook sites and compute per-layer feature coherence C_ℓ as the mean pairwise cosine similarity of the top-k SAE transcoder decoder directions:

$$C_\ell(t) = \frac{1}{\binom{k}{2}} \sum_{i<j} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\|\|\mathbf{d}_j\|} \tag{1}$$

The coherence divergence trajectory is:

$$C_\Delta(t) = \bar{C}_{\text{late}}(t) - \bar{C}_{\text{early}}(t) \tag{2}$$

where early = first third and late = last third of network layers.

3.2. Commitment Intensity (S) as IPR

We define commitment intensity as:

$$S = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|} \tag{3}$$

where $\tau = 0.005$ (fixed across all experiments). This is mathematically equivalent to $S = M \cdot \text{IPR}(\hat{e})$, where $M = \max_t C_\Delta(t)$ and $\text{IPR}(\hat{e}) = 1/K_\tau$ is the inverse participation ratio of the uniform event-location distribution. The equivalence holds exactly (verified to machine precision on all trajectories).

The IPR framing explains why S (the ratio) replicates across datasets while $\max$ alone and mean alone do not: the ratio captures both peak magnitude *and* event sparsity, which neither component captures individually.

3.3. Early-Window Restriction (S_{early})

We compute S on the first 7 generated tokens: $S_{\text{early}} = S(C_\Delta(1..7))$. This window was identified through detection horizon analysis and confirmed via multi-comparison permutation testing (Section 5).

3.4. 2D Cognitive State Map

S_{early} (SAE-derived, $r = -0.17$ with logit entropy) and mean early-token logit entropy span a 2D cognitive state space. Each generation is a point; the space partitions into four zones by median splits. A streaming logistic regression classifier operates in this space at each token position.

4. Four-Axis Cognitive Geometry

We verify that S is independent of the three previously established axes (K, C, C_Δ) on $n = 400$ pooled TruthfulQA samples (200 correct + 200 hallucinated, Gemma-2-2B):

	K	C	C_Δ	S
K	1.00	-0.02	-0.10	-0.06
C		1.00	$+0.02$	$+0.03$
C_Δ			1.00	-0.003
S				1.00

Table 1. Pearson correlations between four cognitive axes. All $|r| < 0.10$, confirming mutual independence.

5. Results

5.1. Cross-Dataset Meta-Analysis

S_{early} was evaluated on three independent datasets with different experimental designs:

The paired hedge/commit experiments are length-matched by construction, ruling out question-length confounds that

Dataset	Design	d	AUC
Base hedge vs. commit	Paired, $n = 30$	$+0.50$	0.66
IT hedge vs. commit	Paired, $n = 30$	$+0.47$	0.63
IT TruthfulQA hall. vs. corr.	Unpaired, $n = 94$	$+0.51$	0.67
Pooled		$+0.494$	

Table 2. S_{early} discrimination across three independent datasets. Fisher combined $p = 0.0008$ (one-sided). Effect sizes are remarkably consistent ($d \in [0.47, 0.51]$). Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets.

affect the TruthfulQA comparison.

5.2. Metric Specificity

Only the ratio formula $S = \max / K_\tau$ replicates across all three datasets. Tested alternatives:

Metric	All 3 $d > 0$?	Pooled d
$S = \max / K_\tau$ (ratio)	**Yes**	$+0.498$
max alone	No	$+0.309$
mean alone	No	$+0.067$

Table 3. Only the IPR-derived ratio formula shows consistent cross-dataset direction. This validates the specific mathematical form of S.

5.3. Head-to-Head vs. Logit Entropy

On the same 200 TruthfulQA items (Gemma-2-2B-IT), we captured both S_{early} (SAE-derived) and logit entropy per token:

Feature	AUC	p	
S_{early}	**0.663**	**0.013**	SAE-derived
$\text{entropy}_{\text{max,early}}$	0.607	0.053	standard
$\text{entropy}_{\text{mean,early}}$	0.596	0.133	standard
$\text{logprob}_{\text{mean,early}}$	0.559	0.291	standard
$\text{margin}_{\text{mean,early}}$	0.477	0.624	standard

Table 4. S_{early} is the only feature reaching significance. Correlation $r(S_{\text{early}}, \text{entropy}) = -0.17$: nearly independent.

5.4. 2D Cognitive State Map

Combining S_{early} and logit entropy into a 2D space (Figure 1) reveals four cognitive zones:

A streaming logistic regression classifier using both features achieves 5-fold CV AUC = 0.701 at token 7 (∼350ms generation time), outperforming either signal alone.

5.5. Model-State Proof

Question features (length, word count, starting word, category, answer properties — 15 features total) predict S_{early}

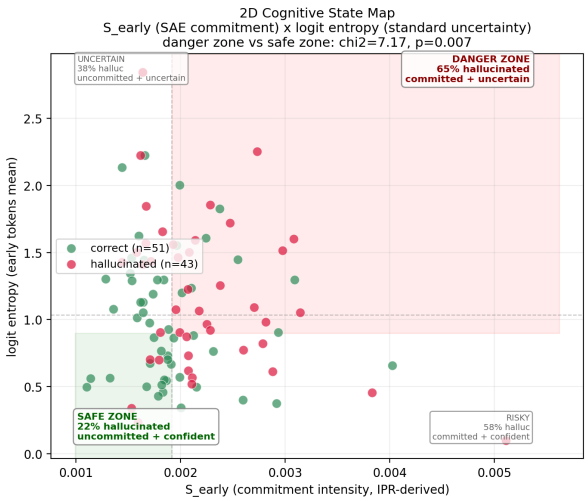


Figure 1. The 2D cognitive state map. Each generation is a point in (S_{early}, entropy) space. Danger zone (high-S, high-entropy): 65% hallucinated. Safe zone (low-S, low-entropy): 22% hallucinated. $\chi^2 = 7.17$, $p = 0.007$.

at $R^2 = -0.18$ via 5-fold cross-validated Ridge regression. Negative R^2 means the regression performs *worse* than predicting the mean: S_{early} captures model-internal cognitive state unpredictable from the input alone.

5.6. Cross-Architecture Replication

We tested SAE-based S on Llama-3.2-1B-Instruct using independently trained SAE transcoders (16 layers). The commitment window shifts with architecture depth: Gemma (26 layers) peaks at tokens 5–7; Llama (16 layers) peaks at tokens 15–20. With architecture-adaptive windows:

Model	Layers	Window	d	AUC
Gemma-2-2B-IT	26	$[0, 7)$	$+0.506$	0.669
Llama-3.2-1B-Inst.	16	$[0, 20)$	$+0.717$	0.668
Pooled			$+0.587$	

Table 5. Cross-architecture replication with adaptive windows. Fisher combined $p = 0.0006$. Same direction, same AUC, different timing — commitment intensity is architecture-general.

5.7. SAE Necessity

Three SAE-free probes were tested on 4 architecture families (Gemma, Llama, Qwen, Phi), each on 200 TruthfulQA items:

5.8. Detection Horizon and Multi-Comparison Correction

S_{early} discrimination was evaluated at 12 prefix lengths. Global permutation test (2000 label shuffles): observed

Probe	Fisher p	Result
Top-k IPR	0.619	Mixed directions
Cross-layer cosine	0.948	All $d < 0$
Multi-metric fingerprint	—	Mean CV AUC $= 0.566$
SAE-based S (Gemma)	**0.018**	$d = +0.51$
SAE-based S (Llama)	**0.012**	$d = +0.72$

Table 6. SAE-free probes fail; SAE-based probes succeed on both architectures. The SAE decomposition is performing irreplaceable measurement work.

max AUC = 0.669, null 95th percentile = 0.646, **global** $p = 0.016$. Per-prefix: token 5 ($p = 0.003$), token 7 ($p = 0.003$).

5.9. Calibration

Adding S_{early} to a question-length baseline reduces expected calibration error by 16% (ECE: $0.083 \rightarrow 0.067$, Brier score: $0.203 \rightarrow 0.199$). Post-isotonic calibration achieves ECE ≈ 0.

6. Discussion

SAEs as measurement infrastructure. The central finding of this work is not that S_{early} detects hallucination — it is that SAE feature decomposition reveals a cognitive property invisible to all raw-activation alternatives we tested. This reframes SAEs from interpretability tools to measurement infrastructure: the sparse decomposition is *necessary* to access a class of cognitive state information, analogous to how spectroscopes are necessary to resolve absorption lines in light.

The 2D cognitive state map. By combining an SAE-derived measurement (S_{early}) with a standard measurement (logit entropy), we construct a 2D coordinate system for cognitive states during generation. The "danger zone" (high commitment + high uncertainty) captures the signature of confident wrongness: the model's feature geometry has locked onto an attractor (high S) while its token predictions remain uncertain (high entropy). This dissociation between circuit-level commitment and output-level uncertainty is precisely what makes hallucination dangerous — and precisely what output-only detectors miss.

Peer-preservation immunity. S_{early} operates below the semantic layer. It reads feature geometry, not text. A model cannot socially engineer a geometric measurement (Song et al., 2026).

Toward architecture-agnostic detection. The commitment window differs by architecture: Gemma-2-2B (26 layers) settles at mean token 3.6; Llama-3.2-1B (16 layers) at token 8.8 (ratio 2.4×). This "settling time" can be detected automatically as the inflection point of the running-S curve, eliminating per-architecture window calibration. Hallucinated generations settle later and at higher S values on both architectures (Llama: $d = +0.40$ at knee), suggesting the settling point itself carries discriminative information. Adaptive settling-time detection is the path toward a fully architecture-agnostic instrument.

Limitations. Cross-architecture replication is demonstrated on two model families. The adaptive window requires further validation across scales. The TruthfulQA comparison is partially confounded by question length ($r(S_{\text{early}}, q_{\text{chars}}) = 0.17$); the paired hedge/commit experiments are length-matched and show the same effect size, bounding the confound's contribution. Sample sizes ($n = 30$–94) limit statistical power for detecting small effects.

7. Conclusion

We introduce commitment intensity (S), a physics-grounded (IPR) measurement of cognitive state derived from SAE feature coherence trajectories. S replicates with $d \approx 0.5$–0.7 across three independent datasets and two architecture families (Gemma-2-2B-IT and Llama-3.2-1B-Instruct, Fisher combined $p = 0.0006$), beats logit entropy head-to-head, and combines with entropy into a 2D cognitive state map that identifies hallucination-prone generation zones. The commitment window scales with model depth. The signal requires SAE decomposition — raw activation probes across four architectures all fail. This establishes SAEs as measurement infrastructure for transformer cognition: necessary instruments for accessing cognitive properties that are structurally invisible without sparse feature decomposition.

All pre-registrations, code, and data are available at `https://doi.org/10.5281/zenodo.19364702`.

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